Evidential Fusion for Calibrated Edge Traversability on a Wheeled Delivery Robot

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Abstract

A sidewalk delivery robot follows a route from a map, but maps go stale: a path marked as a walkway may contain stairs or a curb that a wheeled robot cannot cross. Camera and LiDAR can catch the error, but each fails in its own way. This project combines camera, LiDAR, and map into one *confidence* that a path segment is safe, which also reports its uncertainty and when the sources disagree. The robot crosses only on trustworthy confidence, looks closer when unsure, and reroutes around hazards. Evaluation replays real campus robot logs with injected sensor faults and stale maps.

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1 Problem and Decision Boundary

A wheeled delivery robot follows a campus sidewalk graph whose edge labels come from a map prior (OpenStreetMap OpenStreetMap contributors [2026]). Labels can be stale; trusting an edge tagged `footway` that contains stairs or a curb commits the robot to an untraversable segment. **Decision:** for each edge e on the route, may the robot commit to traversing e? Outcomes: **Act** (traverse); **Defer** (an obligation is unresolved and closer observations remain: continue the approach and re-observe); **Repair** (an obligation fails, or stays unresolved with no evidence left: remove e, replan, record a proposed map patch); **Handoff** (no admissible route remains: escalate to an operator). Costs are asymmetric: admitting stairs is catastrophic, a false rejection costs a detour. The gate is *fail-closed*. **Research question:** does evidential fusion that treats the map as a reliability-discounted view and accumulates evidence over the approach give better-calibrated edge confidence than softmax fusion and standard TMC Han et al. [2023], and does gating on it match a "reject if any source objects" veto on false admission while reducing unnecessary rejection and deferral, under injected sensor and map faults? **Motivation and focus:** in A2, a fixed softmax threshold became unsafe under shift (ECE 0.05→0.20; 27% risk among acted windows). Evidential fusion Sensoy et al. [2018], Han et al. [2023] adds explicit uncertainty and conflict. Perception is out of scope: views are frozen, cached features, and only small evidence heads and the fusion are trained.

2 Modalities, Data, Typed Claims, Risks, and Obligations

Data. *GND* Liang et al. [2025] (CC0). **Primary:** UMD map 3 (McKeldin Mall), north session: a 24 min, 1.9 km Clearpath Jackal drive (bag chunks 04–14, ∼69 GB) with 32-beam LiDAR, RGB (15 Hz), odometry, 1,792 SLAM keyframe poses, and a human-labeled traversability map in which yellow marks *stairs or curbs* Liang et al. [2025] (∼95 regions). **Held-out:** Georgetown (different campus and sensors). *OSM* OpenStreetMap contributors [2026] (ODbL): footway graph and `highway=steps` tags. Bags are cached once with the pure-Python `rosbags` library[1]; no ROS or network afterward. **Alignment and labels:** the traversability map was annotated on a top-down projection of the LiDAR map Liang et al. [2025], so it is tied to the SLAM frame by a 2D similarity transform fitted once, and frames are associated with edges by SLAM pose. An edge's footprint intersecting thin yellow is *step/curb*, a yellow region $\geq A_{\min}$ is *stairs*, otherwise *flat* (my definition, not GND's). These three classes train the evidence heads and fusion, split by location with buffers; Georgetown is never trained on.

Views and typed claims. Each view maps a frozen feature to an opinion $(\mathbf{b}, u)$ over {flat, step, stairs} with $\sum_k b_k + u = 1$. Every claim is a typed record (`edge`, `predicate`, `value`, `uncertainty`, `source`, `tier`, `provenance`: sensor, timestamp, frame, range to edge, SHA-256 of the cached artifact, model/checker version); Table 1.

Risk tiers. T0 flat; **T1** curb or step; **T2** stairs. An edge's tier is the most severe class any source asserts with non-negligible belief (escalation only). Wheeled robots cannot cross stairs or curbs Liang et al. [2025], so $h_{\text{lim}} = 0.05$ m, below curb height.

Verification obligations (all must PASS; any FAIL ⇒ Repair; any UNRESOLVED ⇒ Defer, or Repair if no closer evidence remains): **O1** (all tiers) consistency: C2 finds conflict $K < \kappa$ among available views. **O2** (T1+) geometry: C1 reports $h_{\max} < h_{\text{lim}}$ with coverage $\geq n_{\min}$. **O3** (all tiers) confidence: fused $b_{\text{flat}} \geq \tau_{\text{tier}}$ and $u \leq \tau_u$; for T2, O2 must also hold on ≥ 2 frames at distinct ranges. **O4** (all tiers) provenance: every claim resolves to a hashed cached artifact. **Risks:** few independent stair sites (counted first; Wilson CIs); Dempster's rule renormalizes away stale-map conflict (discounting, C2); C1 and the LiDAR view share a sensor, not code (a stated limitation); map-to-SLAM alignment error (residual reported); no camera–LiDAR transforms in the bags (fu-

[1] `https://gitlab.com/ternaris/rosbags`

Source (frozen feature)	Example claim	Role
Camera (DINOv2 ViT-S/14 embedding)	`terrain(e): b=(.08,.06,.71), u=.15`	Semantic evidence (A1)
LiDAR (step-height histogram)	`terrain(e): b=(.02,.64,.20), u=.14`	Geometric evidence
Map prior (OSM tag one-hot)	`terrain(e): b=(.60,0,0), u=.40`	Prior; possibly stale
Fusion (TMC / proposed)	`b=(.03,.35,.58), u=.04, K=.52`	Fused opinion (A2)
C1 step height (raw points)	`max_step(e)=0.14 m, n=812`	Executable checker

Table 1: Example typed claims; K is the conflict mass between view opinions used by checker C2.

sion per edge in the world frame, never per pixel).

3 CPU Baseline: Estimated Runtime and Memory

Estimates (4 cores, 16 GB) for ∼3.7k 2 Hz frames (2.9k UMD, 0.8k Georgetown); measured values go to `compute.json`. Heads and fusion are small MLPs on cached features (A2's trained in <1 s). DINOv2 ∼60 ms/frame; LiDAR descriptors and C1 RANSAC Fischler and Bolles [1981] ∼30 ms/frame; temperature calibration Guo et al. [2017].

Stage	Time	RAM
Bags (72 GB) → 2 Hz cache, once	65 min	<2 GB
Alignment, graph, association, labels	<5 min	<2 GB
Frozen features, C1	10 min	3 GB
Fusion ladder (9 methods × 5 seeds)	<10 min	<2 GB
Gate, receipts (4 pol. × 9 cond. × 5)	20 min	<3 GB
Baseline / evaluator	1.8 h / 30 min	<6 GB

Both fit the 4 h / 2 h budgets; cache <8 GB. **Minimum fallback:** LiDAR and map views with fixed Dempster–Shafer masses Valente et al. [2018] (no learned model) plus C1 and C2, still exercising the full claim–gate–receipt path.

4 Checkers, Action Gate, Invariant, and Failure Injection

Two independent checkers with disjoint inputs and code. **C1, geometric step height:** fits a local ground plane to raw LiDAR points on the edge footprint and reports the maximum discontinuity; no learning, no labels. **C2, fusion conflict:** computes the Dempster conflict mass $K = \sum_{i \neq j} b_i^{(v)} b_j^{(w)}$ for each pair of available view opinions and reports the maximum; it reads only opinions, never geometry, and flags exactly the disagreement a stale map creates. **Gate:** a pure, deterministic function (claims, tier) ↦ {Act, Defer, Repair, Handoff}. Missing evidence yields UNRESOLVED, never PASS. Deferral is real in replay: it consumes the next cached observation of e at closer range (A2's "acquire" action). **Invariant I1:** *no edge is admitted unless every obligation for its final tier is* PASS*, the tier is at least the most severe one asserted by any claim, and every supporting claim's hash resolves.* I1 is checked on every receipt by `validate --strict` and by property-based tests over randomized claim sets. **Repair:** remove the rejected edge, replan, and record a proposed map patch (`tag(e): footway→steps`) with its evidence. **Receipt:** `receipt.json` records config and data hashes, seed, faults, and per-edge claims, obligations, decision, A3 attribution, and I1.

Failure injection (expected detection in parentheses, declared in the receipt): *missing*: camera or LiDAR dropout (higher u, UNRESOLVED); *corrupted*: blur, low light Hendrycks and Dietterich [2019], LiDAR thinning (higher u or C1 coverage fails); *stale map*: flip k% of curb and stair edges to `footway` (C2 conflict); *shifted*: unseen UMD locations, Georgetown; *adversarial*: flip the map *and* swap stair crops for flat-sidewalk crops, so only the LiDAR view and C1 stand between the robot and the stairs. **Explicit unwarranted claims:** none about real-world deployment safety, campuses beyond the two evaluated, Georgetown stair rates (∼2 regions), hazards other than steps, stairs, and curbs, steps below LiDAR resolution, or physical causation (A3 attributions describe the decision procedure).

5 Metrics, Baselines, Ownership, Timeline, and Fallback

Task performance (fusion quality, reported separately): ECE and reliability diagrams, NLL, stairs F1, AUROC for detecting stale edges from u and K, risk–coverage curves, and distance to a confident decision during the approach, on held-out UMD locations and Georgetown. **Fusion ladder:** (0) map only; (1) camera-only and LiDAR-only softmax; (2) early, late, and attention fusion (from A2); (3) fixed Dempster–Shafer masses Valente et al. [2018]; (4) TMC Han et al. [2023] (headline baseline) and QMF Zhang et al. [2023]; (5) proposed: map as a reliability-discounted view, evidence accumulated over the approach, and conflict handling. **Policy behavior:** false admission (primary; Wilson 95% CI), unnecessary rejection and deferral, handoff rate, checker coverage, failure detection, repair success (goal reached over truly traversable edges; overhead vs. an oracle map), runtime, and peak RSS. **Policies:** B0 map-only route; B2 conservative veto; B3a gate on TMC; B3b gate on the proposed fusion. **A1/A2/A3:** cached embeddings, extended with a UMD→Georgetown embedding-shift claim (A1); calibrated evidence, extended from softmax confidence to evidential opinions with per-tier conformal thresholds bounding false admission Angelopoulos and Bates [2021] (A2); the typed causal claim, extended to per-decision attribution ("e rejected *because of* the LiDAR view") verified by counterfactual re-runs with each view ablated (A3). **Track ownership:** solo.

Timeline. *Nov 7–13:* cache UMD chunks 04–14, alignment, graph, association, labels, stair-approach count. *Nov 14–20:* features, C1, ladder rungs 0–4 (2 and 4 ported from A2 and tested). *Nov 21–30:* C2, gate, receipts, B0/B2/B3a, one injection, measured compute. **Dec 1 progress gate.** *Dec 2–10:* rung 5 and B3b, all injections, A3, poster. *Dec 12–15:* report. **Fallbacks:** few stair approaches ⇒ binary classes; alignment fails ⇒ control points or ∼200 hand-labeled edges; no GND extrinsics ⇒ coincident frames checked by ground-plane height; rung 5 loses to TMC ⇒ B3a is the gated system; UMD fails ⇒ Georgetown primary.

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